



AI & Data Science in Accounting

The Challenge

Students are asked to:

- Develop a conceptual AI/Data Science solution.
- Work with simulated accounting datasets.
- Deliver Executive Summary + Mock-up Presentation.
- Goal: Demonstrate improvements in processes and performance.



Positioning of the Work

Purpose:

- Build demonstrative prototypes with simulated data.
- Apply AI/Data Science to accounting.

Deliverables:

- Executive Summary, Mock-up, Evidence of benefits.

Excluded:

- ERP integration, Enterprise RPA, Industrial compliance.
- Objective: Feasible prototypes at Bachelor level.



Business Cases Index

1. Smart Reconciliation
2. Cash Flow Crystal Ball
3. Fraud Radar
4. Fair Share
5. Margin Watch
6. Closing Guardian
7. Cost Foresight
8. Compliance Copilot
9. Inventory Brain
10. Green Ledger

Cost Foresight – Forecasting Production Costs in Steel

Industry Focus	Steel manufacturing
Problem	Volatile energy/raw material costs make production costs unpredictable.
Application	Predictive model integrates simulated accounting data with market indices to forecast costs.
Deliverables	Executive Summary, forecasting model mock-up, benefits analysis.
Academic Scope	Regression/time series feasible. Excluded: industrial-scale simulations.
Accounting Focus	Apply Gross Margin Trend Analysis and COGS Ratio by cost center. Integrate Inventory Turnover to connect production cost forecasts with asset use.

Key Takeaways & Conclusion

- AI and Data Science can support accounting.
- Focus on demonstrative prototypes with simulated data.
- Deliverables: Executive Summary, mock-up, evidence of benefits.
- Excluded: ERP, RPA, full compliance systems.
- Goal: Conceptual solutions achievable at Bachelor level.

